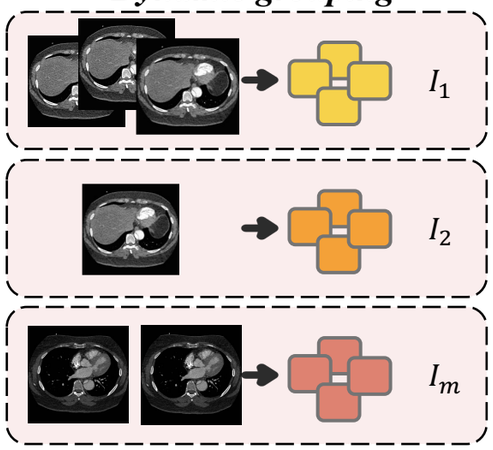


Hyperbolic token composition

1. Ordered support construction

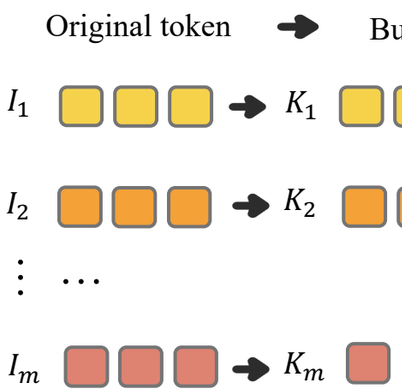
Dynamic grouping



$$g_{m,p}=\frac{1}{|I_m|}\sum_{s\in I_m} x_{s,p}$$

2. Relational allocation

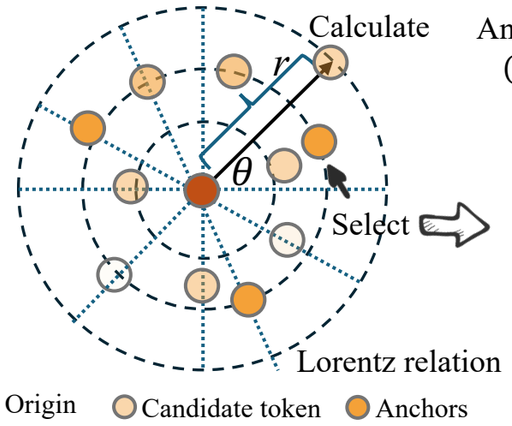
Global representativeness



$$\pi_m=\frac{\exp(\eta_m)}{\sum_n \exp(\eta_n)}, K_m=\min\{b_m, P\}$$

3. Lorentz anchor selection

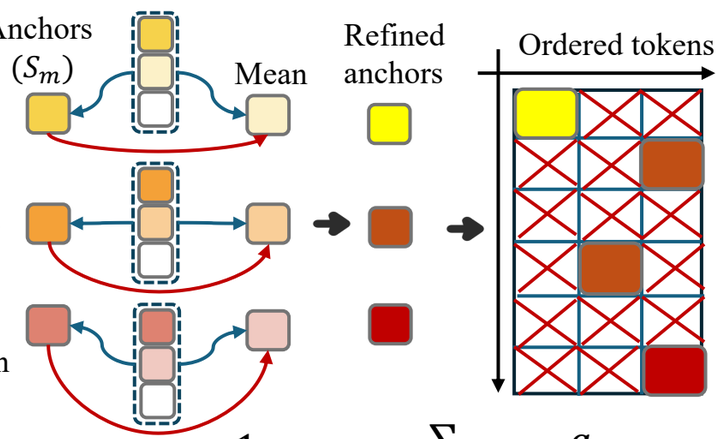
Lorentz local distinctiveness



$$S_m= \mathrm{Top}K_p(u^{\mathbb{H}}_{m,p}, K_m)$$

4. Local refinement & order

Count-preserving fusion



$$\tilde{x}_{m,p}=\frac{1}{2}\left(g_{m,p}+\frac{\sum_{q\in R_{m,p}} g_{m,q}}{\max(1, |R_{m,p}|)}\right)$$

Lorentz distance

$$\cosh d_{\mathbb{H}}(x,y)=\cosh(r_x-r_y)+\sinh r_x \sinh r_y\,(1-\cos\theta_{xy})$$

Inter-unit $d_{\mathbb{H}}(z^a_m, z^a_n)$ for global representativeness

Intra-unit $d_{\mathbb{H}}(g_{m,p}, g_{m,q})$ for local distinctiveness

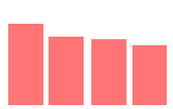
Attention-calibrated capacity control

Focused attention (low entropy)

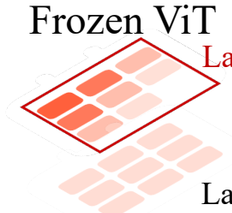
Dispersed attention (high entropy)



Attention distribution



Frozen ViT



Normalized entropy $\bar{H} \in [0,1]$

Attention entropy quantification

Fixed $\rho \in \{10\%, 20\%, 30\%\}$

Calibrated $\rho_{cal} = \mathrm{clip}\left(\frac{\bar{H}^3}{2}; 0, \rho_{max}\right)$

Interface-aware budget determination

Provide budget to composition